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To cite this article: Nathan Walter, Jonathan Cohen, R. Lance Holbert & Yasmin Morag (2019): Fact-Checking: A Meta-Analysis of What Works and for Whom, Political Communication, DOI: [10.1080/10584609.2019.1668894](https://doi.org/10.1080/10584609.2019.1668894)

To link to this article: <https://doi.org/10.1080/10584609.2019.1668894>



Published online: 24 Oct 2019.



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Fact-Checking: A Meta-Analysis of What Works and for Whom

NATHAN WALTER, JONATHAN COHEN,  R. LANCE HOLBERT,
and YASMIN MORAG

Despite its growing prominence in news coverage and public discourse, there is still considerable ambiguity regarding when and how fact-checking affects beliefs. Informed by theories of motivated reasoning and message design, a meta-analytic review was undertaken to examine the effectiveness of fact-checking in correcting political misinformation ($k = 30, N = 20,963$). Fact-checking has a significantly positive overall influence on political beliefs ($d = 0.29$), but the effects gradually weaken when using “truth scales,” refuting only parts of a claim, and fact-checking campaign-related statements. Likewise, the ability to correct political misinformation with fact-checking is substantially attenuated by participants’ preexisting beliefs, ideology, and knowledge. The study concludes with a discussion of the fact-checking literature in light of current gaps and future opportunities.

Keywords meta-analysis, fact-checking, misinformation, correction, motivated reasoning

Political misperceptions are prevalent across contexts and nation-states and have proven themselves to be shockingly resilient to correction (Bode & Vraga, 2015; Garrett, Nisbet, & Lynch, 2013). There are clear normative concerns associated with persistent beliefs grounded in factual inaccuracies (see Althaus, 2012), and perceptions of this kind which appear to strengthen after exposure to correction are especially worrisome (Thorson, 2016). Misinformation, and attempts to correct it, are as old as democracy itself (Grant, 1995), and a wide range of journalistic (e.g. Associated Press) and non-journalistic (e.g., FactCheck.org) organizations currently engage a diverse set of practices and procedures

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to correct misinformation that is perceived to soil the marketplace of ideas (see Young, Jamieson, Poulsen, & Goldring, 2018).

One of the most popular innovations designed to address the prevalence of misinformation has been fact-checking. Fact checking is the practice of systematically publishing assessments of the validity of claims made by public officials and institutions with an explicit attempt to identify whether a claim is factual. News consumers are encountering this practice often across media with an increase of more than 900% in its use since 2001 for newspapers and more than 2,000% in broadcast media (Amazeen, 2013). Often referred to as the successor of ad watches (e.g., Graves, Nyhan, & Reifler, 2016), fact-checkers evaluate the accuracy of claims made by political entities. Among the most recognizable fact-checking organizations are FactCheck.org and PolitiFact. A strong empirical indicator of the pervasiveness of fact-checking by news organizations is that the practice has been embraced by standard-bearers like *The New York Times* and organizations dedicated to fact-checking have won major journalism awards, including a Pulitzer Prize (Graves et al., 2016). This professional routinization has, to some extent, made fact-checking a necessary component of high-quality, comprehensive political coverage (Schumacher-Matos, 2012).

The growing prominence of this element of reporting, however, is not an indication of its ability to correct misinformation. Given the rise of fact-checkers and their successful integration into traditional reporting, one might expect to find mounting evidence pointing to a reduction in the effects of misinformation. However, the empirical evidence regarding the effects of fact-checking is highly divided, with some studies finding that exposure to fact-checking can reduce misinformation (e.g., Fridkin, Kenney, & Wintersieck, 2015), but other works recording null findings (e.g., Garrett & Weeks, 2013), or even boomerang effects (e.g., Nyhan & Reifler, 2010). To address this gap, the present study (1) reports findings from a meta-analysis that examines the overall effectiveness of fact-checking; (2) determines the consistency of the field's empirical findings; (3) tests the role of several theory-driven moderators that are commonly used to explain the contingencies of fact-checking; and (4) identifies areas where additional research is needed.

On Strengths and Limitations of Fact-Checking

The diversity of political topics about which misinformation is published is matched only by the myriad of communication strategies employed in seeking to alter political misperceptions. From simple retractions (e.g., Ecker, Lewandowsky, Swire, & Chang, 2011) and ad watches (e.g., Pfau & Louden, 1994) to full-blown correction campaigns (e.g., Dyer & Kuehl, 1978), different techniques are used to monitor the veracity and correctness of political discourse. The newest members in the family of potential antidotes to misinformation are fact-checking organizations best described as “nonpartisan, nonprofit consumer advocate[s] for voters that aim[s] to reduce the level of deception and confusion in U.S. politics” (FactCheck.org, 2018). Beyond their nonpartisan nature, transparency, and reliance on independent research, these enterprises stand apart from other organizations due to their exclusive focus on factual statements made by major political players, as part of debates, speeches, interviews, and news releases, limiting the topics to claims that can be definitively proven or disproven (Amazeen, 2016). It is not surprising, therefore, that different organizations such as PolitiFact, FactCheck.org, and The Fact Checker generally agree with each other and reach very similar verdicts (Amazeen, 2016; but see Marietta, Barker, & Bowser, 2015).

Though fact-checkers have undoubtedly acquired professional prestige and a large audience (Graves et al., 2016), it is unclear whether they shift people's beliefs and correct misinformation. After all, fact-checkers are outspoken about their primary objective being to inform the public (Amazeen, 2013). To the extent that citizens should desire to have correct information on which to base their beliefs and political attitudes, being told that a claim is incorrect should make them believe it less. This is based on the simple notion that beliefs and attitudes are based on information received from one's environment and that information should affect attitudes (see Wolfsfeld, 2011). The credibility and non-partisan nature of fact-checkers should make their judgment more influential and their pronouncements more effective, all the more so in cases where much conflicting information exists, making it difficult to know who to believe. Despite this expectation of a strong effect of fact-checkers, Zaller (1992) argues that not all information is used to shape beliefs and that people combine portions of available information with preexisting opinions when forming their political beliefs. Indeed, it would be somewhat naïve to suggest that the mere exposure to accurate information can shape beliefs in highly polarized political environments, where the notion of objectivity and impartiality is constantly called into question. But it is also reasonable to expect that accurate information should help shape more accurate beliefs.

When attempting to estimate the efficacy of fact-checking with individual studies, a host of complexities arise. Namely, a close inspection of individual studies reveals contradictory findings, with some researchers summarizing their results as "the failure of fact-checking" (Jarman, 2014, p. 1), whereas others show that "fact-checks influence people's assessments of the accuracy, usefulness, and tone of negative political ads" (Fridkin et al., 2015). This ambiguity is further confounded by the idiosyncratic features of studies that tend to differ along critical dimensions such as individual-differences (e.g., political interest, political ideology) and contextual factors (e.g., political issue, proximity to Election Day). A preferred approach to disentangle such inconsistencies and synthesize research findings is meta-analysis (O'Keefe, 2015). Thus, the first aim of the present study is to synthesize research findings and identify the average effect of fact-checking:

RQ1: What is the average effect of fact-checking on message-consistent beliefs?

With that in mind, a meta-analysis is not necessary to understand that there is no simple main effect of fact-checking. Therefore, the current study also attempts to understand *when* and *how* fact-checking efforts succeed or fail. As has been argued of late in the political communication literature, there is a need to remain steadfast in the study of conditional media effects (see Bennett & Pfetsch, 2018; Neuman, 2016). Hence, the following sections introduce theory-driven moderators that attempt to identify the conditions that either augment or attenuate fact-checking efforts.

Motivated Reasoning

Though traditional theories in education and persuasion identified source characteristics such as credibility (Hovland & Weiss, 1951), expertise (Holvand, Janis, & Kelley, 1953), competence and objectivity (Whitehead, 1968) as prime determinants of message effectiveness, more recent findings have consistently shown that people do not approach messages evenhandedly and that preexisting beliefs play a major role in determining the way information is

processed even in the face of concrete evidence and mounting facts (Flynn, Nyhan, & Reifler, 2017). Motivated reasoning is a dominant theory used to explain the formation and resiliency of political misperceptions (Druckman, 2012; Kahne & Bowyer, 2017). This perspective argues that citizens are more accepting of (mis)information that match their pre-existing worldview. Furthermore, the ease with which misinformation can be placed within a pre-existing worldview makes it all the more difficult to correct (Lewandowsky, Ecker, Seifert, Schwarz, & Cook, 2012). Often traced back to Maslow's hierarchy of needs, the theory argues that people are guided by different goals when they process information, including directional goals (i.e., confirming a preexisting belief) and accuracy goals (i.e., trying to be correct and precise; Kunda, 1990; Nir, 2011).

Unsurprisingly, in the political context, directional goals appear to be a very common way to process information (Taber, Cann, & Kucsova, 2009). For instance, voters are likely to feel dissonance if they supported a culturally-conservative candidate and then later read a fact-checking report accusing the candidate of infidelity. Given the psychological discomfort that will arise from trusting the revealing news report, individuals will be motivated to reduce their ambivalence by carefully scrutinizing the new information in an attempt to discredit the report in a manner that will allow for continued support for the candidate. In such situations, people are likely to point to a political bias of the fact-checking organization, the reliance of the report on anonymous sources, or a secret agenda to damage the popularity of the candidate.

Notably, though the frameworks of motivated reasoning and cognitive dissonance have gained prominence in the fact-checking literature, it is important to note that other theories can be used to explain people's tendency to stick to their preexisting beliefs even in light of reliable corrections. For instance, the mental models approach argues that individuals construct coherent representations of unfolding events (Johnson & Seifert, 1994). If a fact-checking message dismisses key information in the established mental model, individuals will be reluctant to accept the new information because it will result in an incomplete or incoherent representation (Lewandowsky et al., 2012). In line with this logic, individuals with a mental model that identifies the free market system as a way to reach the highest standard of living will be reluctant to accept a fact-checking message that speaks to the benefits of regulating the U.S. healthcare system not merely because it contradicts their political ideology (i.e., motivated reasoning) but also due to its ability to create gaps in an otherwise coherent mental model. In sum, the literature proposes several cognitive mechanisms to explain how individuals can prioritize favorable or coherent information even if it is inaccurate or unreliable. Based on this, it was hypothesized that:

H1: The effect of fact-checking on beliefs is stronger for pro-attitudinal corrections compared to counter-attitudinal corrections.

Studies have demonstrated that motivated reasoning affects some news consumers more than others. For example, politically-sophisticated individuals – who also tend to be more partisan – are likely to scrutinize (i.e., counter-argue) fact-checking more than their less knowledgeable counterparts (Young et al., 2018) and thus be less likely to accept corrections of misinformation. For instance, while studies found that cognitive reflection (i.e., the propensity to engage in analytical reasoning) is negatively correlated with adoption of fake news and positively correlated with the ability to discern fake news from real news (Pennycook & Rand, 2017), Nyhan, Reifler, and Ubel (2013) demonstrated that attempts to correct misinformation backfired among knowledgeable conservatives. Similar evidence of

motivated reasoning among political sophisticates demonstrated that climate change knowledge among conservatives decreased with greater attention to political news (Nisbet, Cooper, & Ellithorpe, 2015). It is, therefore, hypothesized that:

H2: The more sophisticated the individual, the less their beliefs will be affected by exposure to fact-checking.

Further, though motivated reasoning is evident among both Republicans and Democrats, there is some evidence to suggest that the former partisan group might be especially susceptible to its bias in relation to current fact-checking practices. First, Republicans consistently report more negative perceptions of mainstream media than Democrats (Swift, 2016). Second, Democrats fare much better than Republicans in recent fact-checking, with statements made by Republicans more likely to be classified as false (Amazeen, 2013; Wintersieck, 2015). Third, popular conservative media outlets such as *Fox News* and *Breitbart* have been outspoken in their disdain of fact-checking organizations, referring to them as “leftists” and “partisan” (Bokhari, 2018). Fourth, studies have demonstrated that conservatives are more prone to closed-mindedness (Levitan & Visser, 2008) and less likely to seek-out attitude-dissonant information (Barberá, Jost, Nagler, Tucker, & Bonneau, 2015). Finally, while Democratic presidential candidates have embraced fact-checking organizations, their Republican counterparts - in the 2012 and the 2016 presidential elections – vocally condemned fact-checkers as biased (Shin & Thorson, 2017). Based on these collective insights, the following is posited:

H3a: The effect of fact-checking on beliefs is stronger for Democrats/liberals compared to Republicans/conservatives.

H3b: There is an interaction effect such that Republicans/conservatives are more likely to accept pro-attitudinal fact-checking and less likely to accept counter-attitudinal fact-checking, compared with Democrats/liberals.

The level of motivated reasoning may also vary by the context in which fact-checking takes place. It is not surprising that people are more likely to be in-tune with political matters and more likely to be exposed to misinformation during an election season (Lewandowsky et al., 2012). Hence, the confluence of opportunities for, and challenges to, fact-checking is at its peak during election events (Coddington, Molyneux, & Lawrence, 2014). Though there is an obvious opportunity to educate a larger and more attentive audience during election season, motivated reasoning is especially prevalent during these times. As argued by Miller and Conover (2015), “average partisans in contemporary U.S. politics view elections as group competitions in which partisan identities are at stake” (2015, p. 225). Hence, as elections near, the processing of political information tends to rely less on facts and more on political calculations and the interest of the political party (Achen & Bartels, 2017; Kunda, 1990). In line with these findings, it is hypothesized that:

H4: The effect of fact-checking on beliefs is weaker when it concerns election candidates/campaign statements than routine fact-checking.

(In)effective Presentation

Though motivated reasoning is the dominant theoretical approach to explaining the effectiveness of fact-checking, another body of knowledge links the success and failure to correct misinformation to a range of message variables. Accordingly, fact-checking may fail not merely because it deals with value-laden beliefs but simply because it is presented ineffectively. One of the prominent features of fact-checking that differentiates it from other approaches to correction of misinformation is its deployment of graphical meters or “truth scales.” From Politifact’s Truth-O-Meter to *The Washington Post*’s counting illustrations of pinocchios, fact-checkers proved to be remarkably innovative in supplementing their text with visual representations. Given the complex nature of political information, visual rating scales that provide a simplified summary of the fact-checking message (e.g., mostly true, completely false) may facilitate the correction of misinformation, especially among less engaged individuals (Amazeen, Thorson, Muddiman, & Graves, 2018). Put differently, presenting corrective information in graphical forms can serve as a heuristic cue and increase its acceptance compared with messages that rely solely on text (Nyhan & Reifler, 2012). Hence, we pose the following hypothesis:

H5: The effect of fact-checking on beliefs is stronger when it integrates a “truth scale”.

Relatedly, message length can serve as an additional presentation factor that potentially influences fact-checking. In line with dual-processing models (e.g., Petty & Cacioppo, 1986), message length is considered a heuristic cue that enables a quick judgment regarding the veracity of the argument - longer messages are often believed to be more persuading because they, ostensibly, contain more arguments. As indicated in a meta-analysis of involvement and persuasion (Johnson & Eagly, 1989), the length of the message increases persuasion for less-involved individuals. Similarly, Shen, Sheer, and Li (2015) found that message length is significantly correlated with narrative persuasion, such that, on average, longer narratives tend to be more persuasive. By the same token, longer fact-checking messaging may also enjoy this heuristic benefit. Therefore, the following hypothesis is proposed:

H6: The longer the fact-checking message, the more effective it is in influencing message-related beliefs.

Another message characteristic that can play a substantial role as a heuristic or a systematic cue is lexical complexity (Chaiken & Eagly, 1976). As with other message features, fact-checking organizations are expected to navigate a tricky terrain, discussing complex and often esoteric issues while simultaneously attempting to effectively translate the information to the general public. Thus, it is not surprising that FactCheck.org is often criticized for using convoluted language and lacking simplicity (Media Bias/Fact Check, 2016). Complex language can be perceived as elitist potentially alienating audiences. Yet, though simple language can make fact-checking more accessible, it can also compromise its perceived accuracy and objectivity for politically-sophisticated audiences (Shulman & Sweitzer, 2018). Therefore, the following hypotheses are proposed:

H7a: The more complex the fact-checking message, the less likely it is to be effective in influencing message-related beliefs.

H7b: There is an interaction effect such that, lexical complexity carries less influence for politically-sophisticated individuals.

Finally, a more recent line of research suggests that fact-checking messages can backfire due to an “implied truth effect” (Pennycook & Rand, 2017). The logic that underlies this concern is simple: (1) not all political statements are factually verifiable, as a large part of political talk focuses on opinions (Amazeen, 2016); (2) in general, people are bad at distinguishing between statements that can be factually verified and statements that represent personal views (Pew Research Center, 2018); (3) even if a fact-checker’s verdict is that a particular statement was false, readers may infer that if the remaining information was not explicitly tagged as *false*, it must be true. According to this naïve theory, the implication of the *absence* of a disconfirmation is ambiguous: does it imply that the information is verified (leading to the unintended “implied truth effect”) or does it mean that the information was not fact-checked (Pennycook & Rand, 2017). Simply put, in an environment where some statements are flagged as inaccurate and others are not, individuals may assume that unflagged messages are trustworthy (Pennycook & Rand, 2017). To this end, compared to messages that receive an ambiguous judgment (e.g., mostly false), messages that are flagged as completely false are more likely to be corrected, since equivocal verdicts may encourage individuals to wrongfully trust portions of the message that were not explicitly flagged as false. Thus, the final hypothesis is posed:

H8: The effect of fact-checking on beliefs is stronger for fact-checking that refutes the entire message rather than fact-checking that refutes only portions of a message.

Method

Selection Criteria

Literature Search. Five different strategies were used to generate effect sizes. First, electronic databases (e.g., *Google Scholar*, *All Academic*, *JSTOR*, *Medline*, *ProQuest*, *PubMed*, *Communication and Mass Media Complete*, *Educational Resources Information Center*, *PsycINFO*) were searched using relevant key terms (and their derivations), including “fact-check,” “misinformation,” “disinformation,” “correction,” and “continued influence.” Those key terms were often coupled with research outcomes such as “beliefs,” “attitudes,” “perceptions,” “intent,” and “behavior.” This initial search yielded potentially relevant journal articles, books, dissertations, and conference papers that were screened for retrieval on the basis of their title or abstract. Second, reference lists were examined and additional relevant studies (albeit, with many duplicates) were retrieved. Third, recent programs of leading conferences in communication and political science (i.e., APSA, ICA, NCA) were systematically searched using the same key terms. Fourth, we shared a brief explanation of this project along with its inclusion criteria on two large list servers to solicit further relevant studies. Finally, based on personal acquaintance and the list of authors from our dataset, 17 leading scholars in the field of misinformation, journalism, and political communication were contacted and asked to review our corpus (both included and excluded studies) and identify omissions. In total, 11 individuals replied, providing us with four additional studies, which were later added to the dataset.

Inclusion Criteria. Studies were retained based on five criteria: (1) research reports had to include an experimental comparison between exposure to fact-checking and a completely equivalent condition that was not exposed to fact-checking (e.g., Amazeen et al., 2018) or a pretest-posttest experiment where outcomes were assessed before and after exposure to fact-checking (e.g., Jarman, 2016); thus, isolating the effect of exposure to a fact-checker;¹ (2) fact-checking message had to discredit an original statement made by a political entity;² (3) study material had to expressly name either a real or a fictitious fact-checking organization; (4) research reports had to provide a quantitative estimate for the effect of fact-checking on message-relevant beliefs/attitudes; (5) studies had to report on relevant statistics (e.g., means, *SDs*, *t*-values, exact-*p*, Cohen's *d*, odds ratio, frequencies, zero-order correlations). When appropriate statistical data was missing ($k = 4$), we attempted to obtain the information from the corresponding authors (two relevant studies were excluded because the relevant information could not be obtained). As illustrated in Figure 1, the final sample included 30 individual studies (14, 46.67% unpublished), from 20 research reports, with a total sample size of 20,963 ($M = 698.77$, $Med = 465.5$, $SD = 594.39$).

Coding of Variables

Outcomes. Single effect size of exposure to fact-checking on message-consistent beliefs was calculated per sample. Though there is no single consistent and accepted definition for attitudes and beliefs across different disciplines, following Rokeach's (1968) explication, we identified beliefs as a specific judgment of true/false, good/bad, or desirable/undesirable toward an object or a situation; whereas attitudes were conceptualized as an organization of several favorable or unfavorable beliefs. Hence, keeping in mind the overlap with which beliefs and attitudes are used in the literature, there is good rationale to combine them into a single index, which is the way the results will be presented.

It can be argued that different types of beliefs should not be synthesized and aggregated since issue favorability is not the same as issue accuracy, beliefs about immigration are different from beliefs about healthcare, and fact-checking of a politician is not equivalent to rating of information originating from a news outlet. Despite such potential contradictions, the current analysis synthesizes different belief domains for several reasons. First, this practice is in line with other meta-analyses in the area of misinformation (i.e., Chan et al., 2018; Walter & Murphy, 2018), as well as political communication more broadly (e.g., Benoit, Hansen, & Verser, 2003; Boulianne, 2009; Paul, Salwen, & Dupagne, 2000) that aggregated and averaged outcomes across different topics. In fact, compared with other studies of misinformation that averaged beliefs across diverse contexts such as politics, health, marketing, and science, with a wide range of correction methods, including fact-checking, retractions, and forewarnings (i.e., Chan et al., 2017; Walter & Murphy, 2018), the current study seems to be on safer grounds as it has a narrower focus on fact-checking of political statements. Second, from an empirical point of view, potential variations among different types of beliefs made no significant difference in terms of their effect size. Of note, the average effect of fact-checking on beliefs did not vary as a function of question type ($Q(1) = 1.75$, $p = .19$),³ policy domain ($Q(3) = 5.69$, $p = .14$),⁴ issue type ($Q(1) = 0.21$, $p = .65$),⁵ and source of misinformation ($Q(2) = 3.24$, $p = .20$).⁶

The vast majority of studies reported a single effect of fact-checking on beliefs; yet, in five cases, two or more different estimates were provided (e.g., Weeks, 2015; belief in claims

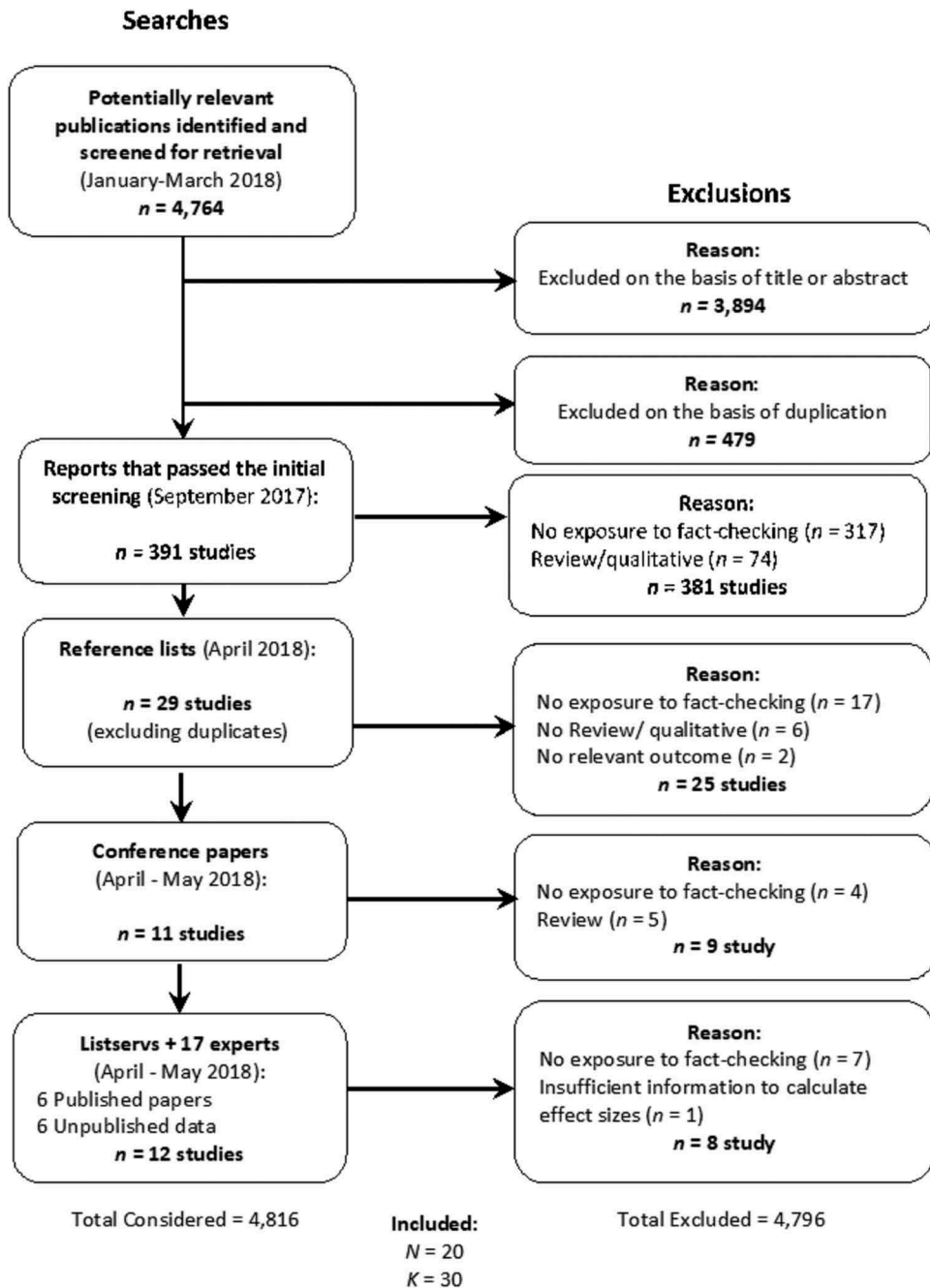


Figure 1. Search strategy.

Note. Only completely irrelevant entries were excluded on the basis of the title/abstract, including patents and newspaper articles. All academic papers that were considered to be even remotely relevant were further examined.

about immigration/beliefs in claims about the death penalty). When more than one effect size was reported, estimates were averaged to represent a single effect size of change in the direction advocated by the fact-checking message. Further, due to concerns over generalizability, when studies included different conditions of exposure to fact-checking messages, only those that corresponded with real-world practices were retrieved for the meta-analysis. For example, we focused on traditional fact-checking as opposed to a humorous corrective video (e.g., Young et al., 2018), fact-checking messages that integrate partisan cues (e.g., Wintersieck, 2015), and an aggregated summary of fact-checking data (e.g., Agadjanian et al., 2019). Likewise, though we included studies that used real-world fact-checkers (e.g., FactCheck.org) and studies that used fictitious fact-checkers (e.g., GetTheFacts.org), the inclusion criteria required studies to frame the source of the corrective message as a fact-checker. Hence, studies that used corrective information that was not explicitly framed as coming from a fact-checker were excluded (e.g., Bode & Vraga, 2015; Wood & Porter, 2019). Considering the fact that only a small fraction of the misinformation literature deals with fact-checking, by including all types of corrections, it would have been impossible to discern findings that are relevant to fact-checking from findings that speak more broadly to misinformation and the continued influence effect (Chan, Jones, Hall Jamieson, & Albarracín, 2017; Walter & Murphy, 2018).

Moderators. Ten theoretically-driven variables were tested. Broadly speaking, these moderators reflect two frameworks that explain the contingencies of fact-checking - motivated reasoning and message presentation. To protect against violations of independence of effect sizes, most moderators were coded at the level of the study, treating a study as the unit of analysis (for an explanation see Borenstein, Hedges, Higgins, & Rothstein, 2009; Hunter & Schmidt, 2004). When moderators were either manipulated or tested within studies (i.e., pro/counter-attitudinal fact-checking, participants' political affiliation, and integration of visual "truth scales"), subsamples were treated as the level of analysis.⁷ In cases where the original study did not provide the complete version of the fact-checking message ($k = 5$), information was obtained from corresponding authors.

Seventeen effect sizes were associated with pro-attitudinal fact-checking (e.g., fact-checking of false negative claims about Obamacare among Democrats) and 18 were associated with counter-attitudinal fact-checking (e.g., fact-checking of false negative claims about the Keystone Pipeline among Republicans). Political sophistication was measured as a continuous variable in the six studies that directly assessed participants' political knowledge and then standardized from 0 to 1 ($M = .56$, $SD = .16$). Overall, 20 studies attempted to use fact-checking to correct campaign-related misinformation compared with 10 studies that focused on routine fact-checking. With respect to visual components of fact-checkers, eight studies used fact-checks that integrated truth scales or visual indicators and 23 samples were exposed only to text. The average word count for a fact-checking message was 307.22 ($SD = 365.19$). A validated lexical complexity analyzer (LCA) was used to gauge the complexity of fact-checking messages. Though there are numerous ways to assess lexical complexity, it is commonly treated as a multidimensional construct, consisting of lexical sophistication, variation, density, and vocabulary errors (Read, 2000). Keeping in mind that the fact-checking messages used in the primary studies were relatively short and screened for grammatical errors, textual density and vocabulary errors were considered less relevant. To this end, lexical complexity was gauged with four different indices, including lexical sophistication ($M = .37$,

$SD = .11$), lexical word variation ($M = .71$, $SD = .13$), verb variation ($M = .18$, $SD = .12$), and noun variation ($M = .66$, $SD = .13$) (for more information see Ai & Lu, 2010; Lu, 2012).⁸ Then, the four different indicators were averaged to create a single composite of lexical complexity ($M = .47$, $SD = .09$, $\alpha = .82$). Finally, 15 fact-checkers refuted an entire original statement and 15 refuted portions of a statement. Table 1 provides a summary of the studies included in the meta-analysis.

Inter-Coder Reliability. The first author trained an independent coder on a subsample of related studies that were not included in the final sample. Then, reliability was calculated on 41.9% of the total database using Krippendorff's alpha. The overall reliability was satisfactory, with the agreement ranging from .81 to 1.00. Disagreements were resolved by discussion and further guidelines were provided.

Data Analysis

All analyses were conducted using the statistical package Comprehensive Meta-Analysis (CMA) (version.3; Borenstein, Hedges, Higgins, & Rothstein, 2005). The effect size estimate employed was Cohen's d , allowing easy interpretation of both directionality and strength. The results are based on uncorrected estimates of random-effects models (Hedges & Vevea, 1998). As argued by Card (2012), meta-analyses using fixed-effects models are only justified in drawing conclusions about the specific set of studies included in the sample, whereas random-effects models allow to generalize beyond the particular set of studies to a broader population of potential research. In CMA, random-effects models are computed by assigning more weight to the studies that carry more information, based on the inverse of a study's variance (Borenstein et al., 2005).

After assessing the average effect of fact-checking on message-consistent beliefs, heterogeneity was assessed using the Q statistics. In such models, a significant Q coefficient represents data heterogeneity beyond that expected by sampling error alone (Higgins & Thompson, 2002). Then, moderator analyses were conducted to explore potential causes of heterogeneity (Ashford, Edmunds, & French, 2010). Specifically, for categorical variables (i.e., $H1$, $H3a$, $H4$, $H5$, $H8$), moderation analyses focused on Q statistics, by comparing the mean variability in effect size estimates across different values of the moderator. For continuous variables (i.e., $H2$, $H6a$, $H7a$) and interactions (i.e., $H3b$, $H6b$, $H7b$), a meta-regression was employed, allowing to test different moderators as potential predictors of effect sizes. Due to concerns over statistical power, all moderation analyses included, at least, five cases for each value (for a discussion of statistical power in meta-analysis see Jackson & Turner, 2017).

Results

Effects of Fact-Checking

The first research question dealt with the average effect of fact-checking on beliefs. Across 30 studies, the mean effect size of fact-checking was positive and significant ($d = 0.29$, 95% CI [.23, .36], $p = .005$), with significant heterogeneity in effect sizes, $Q(29) = 207.83$, $I^2 = 86.05\%$, $p = .0005$ (see Figure 2 for a forest plot of effect sizes by study). The heterogeneity of effect sizes provides further evidence that the influence of fact-checking is potentially contingent on various moderating variables.

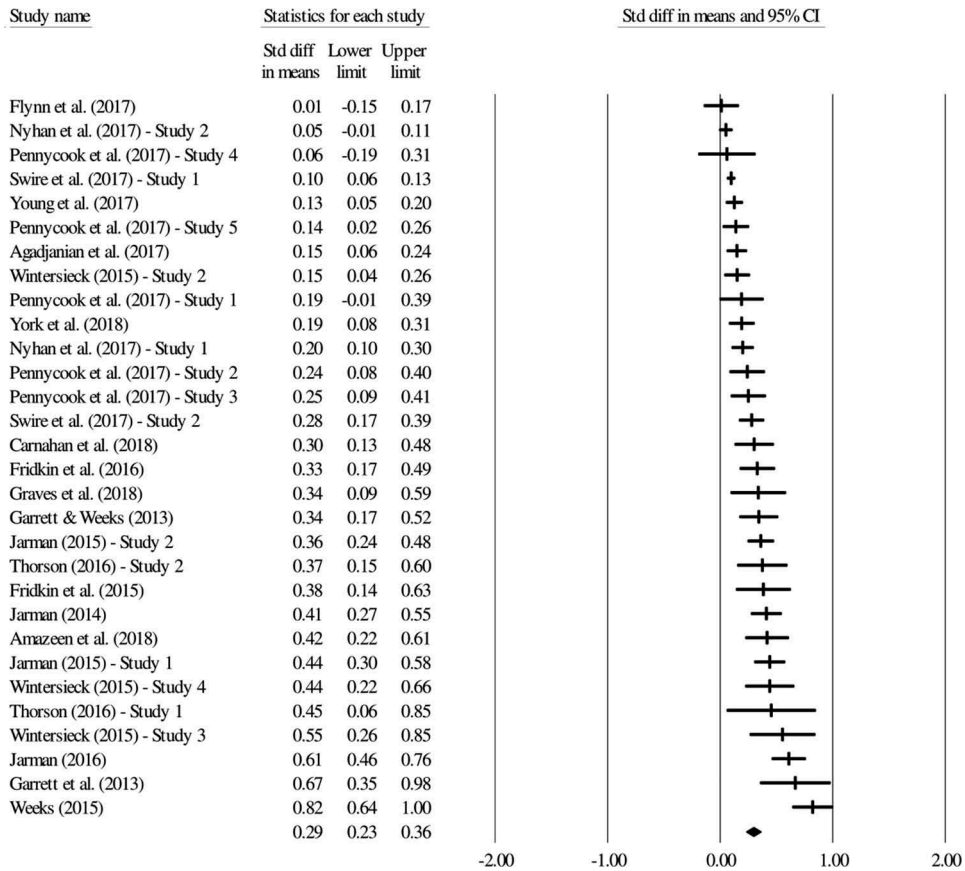


Figure 2. Forest plot for the effect (Cohen's d) of fact-checking on beliefs compared with a neutral control condition.

Note. The effects presented in the forest plot concern the full study samples.

Fact-Checking and Motivated Reasoning

As predicted by H1, there was a significant difference between counter-attitudinal fact-checking and pro-attitudinal fact-checking ($Q(1) = 3.41, p = .04$). In particular, pro-attitudinal fact-checking (i.e., debunking the opposing ideology) resulted in stronger effects ($d = 0.43, 95\% \text{ CI } [.29, .53], p = .001, k = 17$) compared with counter-attitudinal fact-checking (i.e., debunking personal ideology) ($d = 0.28, 95\% \text{ CI } [.19, .38], p = .001, k = 18$). Additionally, as predicted by H2, the efficacy of fact-checking tended to weaken with increased political sophistication; $b = -1.18, SE = .53, (Q(1) = 5.04, p = .02)$. The analysis also tested a potential curvilinear relationship between political sophistication and fact-checking. Yet, the data failed to find any support for this assumption; $b = 1.30, SE = 2.89, p = .65 (Q(2) = 5.48, p = .06)$.

Showing no support for H3a, the analysis did not find a significant difference ($Q(1) = 0.69, p = .41$) between samples who self-identified as Democrats/liberals ($d = 0.31, 95\% \text{ CI } [.20, .42], p = .001, k = 17$) and samples who self-identified as Republicans/conservatives ($d = 0.27, 95\% \text{ CI } [.16, .37], p = .001, k = 17$). To further probe the role

Table 1
List of studies included in meta-analysis

Study	Year	Study Number	N	Lexical Complexity ^a	Political Sophistication ^r	Topic	Fact-Checker
Agadjanian et al. ^{cdefglp}	2019	1	1882	-	-	False statement by a Senator (McConnell/Reid)	PolitiFact
Amazeen et al. ^{abcdegklo}	2018	1	677	.33	-	False statement by a fictitious Congressman	Fictitious
Camahan et al. ^{bhmo}	2018	1	507	.37	-	Misinformation about Common Core policy	Fictitious
Flynn et al. ^{adefhjnp}	2017	1	648	.55	-	Fraud allegations against a legislator	Fictitious
Fridkin et al. ^{adehkmo}	2015	1	452	.48	.55	Negative ads during a campaign	Fictitious
Fridkin et al. ^{ahkmo}	2016	1	452	.48	-	Negative ads during a campaign	Fictitious
Garrett & Weeks ^{befhjno}	2013	1	574	.31	.60	Electronic health records	Factcheck
Garrett et al. ^{hjo}	2013	1	300	.39	.34	Islamic Cultural Center in New York City	Factcheck
Graves et al. ^{bdefgkmp}	2018	1	1876	.41	-	Gun Rights debate	PolitiFact
Jarman ^{dehmo}	2014	1	217	.41	-	Gun Rights debate	Fictitious
Jarman ^{adehlp}	2015	1	187	.38	-	Obamacare	Fictitious
Jarman ^{adehlp}	2015	2	269	.37	-	Obamacare	Fictitious
Jarman ^{bdehlp}	2016	1	198	.44	-	Immigration Policy	Fictitious
Nyhan et al. ^{abdefhlo}	2017	1	1680	.51	-	Crime Rates	Fictitious
Nyhan et al. ^{abdefhlo}	2017	2	527	.51	-	Crime Rates	Fictitious
Pennycook et al. ^{abefijno}	2017	1	479	.55	-	Claims about a candidate	Fictitious

(Continued)

Table 1
(Continued)

Study	Year	Study Number	N	Lexical Complexity ^a	Political Sophistication ^r	Topic	Fact-Checker
Pennycook et al. ^{abcfijno}	2017	2	1879	.55	-	Claims about a candidate	Fictitious
Pennycook et al. ^{abcfijno}	2017	3	1495	.55	-	Claims about a candidate	Fictitious
Pennycook et al. ^{abcfijno}	2017	4	400	.55	-	Claims about a candidate	Fictitious
Pennycook et al. ^{abcfijno}	2017	5	1018	.55	-	Claims about a candidate	Fictitious
Swire et al. ^{adehlo}	2017	1	1776	.61	-	Statements about vaccines	Fictitious
Swire et al. ^{adehlo}	2017	2	320	.61	-	Statements about vaccines	Fictitious
Thorson ^{adhnp}	2016	1	122	.59	-	Accusations of misconduct	Fictitious
Thorson ^{adehnp}	2016	2	345	.51	-	Accusations of misconduct	Fictitious
Weeks ^{dehjklo}	2015	1	512	.40	.50	Death penalty/immigration	AP
Wintersieck ^{abfhkmo}	2015	2	452	.48	.55	Negative ads during a campaign	Fictitious
Wintersieck ^{abdefhklp}	2015	3	134	.48	-	Political debate	PolitiFact
Wintersieck ^{abfhklp}	2015	4	284	.48	-	Negative ads during a campaign	PolitiFact
York et al. ^{fhkno}	2018	1	1190	.59	-	Mandatory immunization	Fictitious
Young et al. ^{dehiklo}	2018	1	111	.32	.85	Keystone Pipeline	FactCheck

Note. ^aCampaign-related; ^bpartial refutation; ^cinclusion of graphical elements; ^dcounterattitudinal correction; ^eproattitudinal correction; ^funpublished; ^garticle +rating; ^harticle only; ⁱlabel only; ^jreference to objectivity; ^klogo; ^lpolitician as source; ^minterest group as source; ⁿpress as source; ^obelief accuracy; ^pbelief favorability; ^qcomposite score based on four different indicators of lexical complexity, including sophistication, lexical word variation, verb variation, and noun variation; ^rstandardized score (0–1) based on studies that measured political knowledge.

played by ideology, we examined the interplay between fact-checking messages and participants' political affiliation. Interestingly, when focusing only on Democratic/liberal participants, the recorded difference between fact-checking of counter-attitudinal and pro-attitudinal messages disappeared ($Q(1) = 0.93, p = .34$). Simply put, for Democrats/liberals, there was no significant difference between counter-attitudinal fact-checking ($d = 0.32, 95\% \text{ CI } [.17, .47], p = .005, k = 6$) and pro-attitudinal fact-checking ($d = 0.43, 95\% \text{ CI } [.26, .61], p = .001, k = 7$). Yet, for Republicans/conservatives ($Q(1) = 9.55, p = .001$), there was a significant difference in favor of pro-attitudinal corrections ($d = 0.52, 95\% \text{ CI } [.32, .73], p = .001, k = 5$), compared with counter-attitudinal corrections ($d = 0.17, 95\% \text{ CI } [.07, .26], p = .003, k = 8$). In other words, and supporting H3b, Republicans/conservatives appear to be more susceptible to motivated processing compared to their Democratic/liberal counterparts. Figure 3 illustrates the interaction between fact-checking valence and political ideology. Additionally, as hypothesized in H4, fact-checking effects were significantly weaker for campaign-related statements ($d = 0.24, 95\% \text{ CI } [.17, .30], p = .005, k = 20$), as opposed to non-campaign fact-checking ($d = 0.38, 95\% \text{ CI } [.26, .51], p = .005, k = 10; Q(1) = 4.07, p = .04$).

Fact-Checking and Message Design

With respect to fact-checking design characteristics, a pertinent question deals with the inclusion of truth scales or graphical elements that can potentially facilitate fact-checking. The treatment of graphical elements as a moderator, resulted in a significant difference between fact-checkers that included visual accuracy cues and those that did not ($Q(1) = 6.58, p = .01$). Specifically, and contrary to H5, the inclusion of graphical elements appears to backfire and attenuate correction of misinformation, with visual fact-checkers producing significantly weaker effects ($d = 0.19, 95\% \text{ CI } [.14, .24], p = .001, k = 8$), as

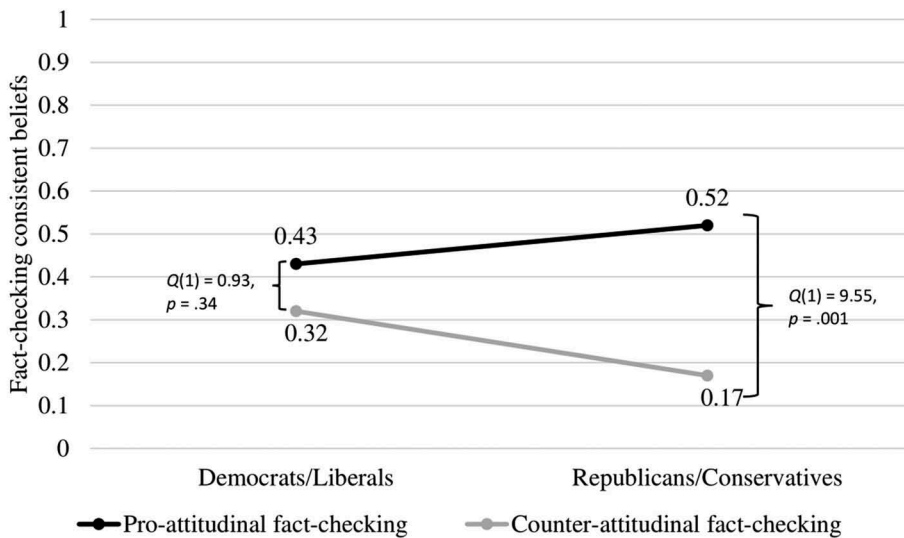


Figure 3. The interaction effect of fact-checking on beliefs by political affiliation/ideology and pre-existing beliefs.

opposed to fact-checking that does not include visual accuracy cues ($d = 0.31$, 95% CI [.23, .39], $p = .001$, $k = 23$).

Contrary to H6, there was no significant effect of message length on the efficacy of fact-checking; $b = .01$, $SE = .02$, ($Q(1) = 0.30$, $p = .58$). The role played by the lexical complexity of the fact-checking message was also tested with a meta-regression. Effect sizes associated with fact-checking were treated as the outcome and message length (i.e., number of words) and lexical complexity, a composite of lexical sophistication, lexical word variation, verb variation, and noun variation, were entered as predictors. The results of the meta-regression indicated that lexical complexity is a significant negative predictor of fact-checking efficacy; $b = -1.43$, $SE = .43$, $p = .001$ ($Q(1) = 12.17$, $p = .001$). Simply put, and in line with H7a, linguistic complexity had a taxing effect on fact-checking, so that more simplistic messages tended to achieve more desirable results. Further, there was no support for H7b as no significant interaction between lexical complexity and political sophistication was found; $b = -6.64$, $SE = 27.87$, $p = .81$; ($Q(3) = 9.25$, $p = .02$). Likewise, as a post-hoc test, we assessed whether the effects of lexical complexity on fact-checking follow a curvilinear pattern. The results did not support a potential curvilinear effect; $b = -3.75$, $SE = 4.58$, $p = .41$ ($Q(3) = 12.68$, $p = .005$). Finally, the analysis found support for the implied truth effect (H8; $Q(1) = 4.30$, $p = .03$). In particular, fact-checkers that attempted to refute an entire statement ($d = 0.31$, 95% CI [.23, .40], $p = .001$, $k = 15$) were substantially more effective than fact-checkers that sought to correct only parts of a statement ($d = 0.19$, 95% CI [.12, .27], $p = .001$, $k = 15$).⁹

Stimulus Characteristics

Finally, the analysis focused on several critical dimensions of fact-checking messages. In particular, there was a significant difference ($Q(2) = 6.66$, $p = .04$), between messages that included only fact-checking labels ($d = 0.18$, 95% CI [.11, .26], $p = .001$, $k = 5$), fact-checking articles ($d = 0.32$, 95% CI [.24, .40], $p = .001$, $k = 22$), and fact-checking articles with rating scales ($d = 0.28$, 95% CI [.10, .47], $p = .003$, $k = 3$). Further, there were no difference in effect sizes ($Q(1) = 0.83$, $p = .36$) between messages that targeted specific political actors ($d = 0.27$, 95% CI [.21, .34], $p = .001$, $k = 23$) as opposed to general false claims ($d = 0.36$, 95% CI [.19, .52], $p = .001$, $k = 7$). In addition, there was no significant difference between messages that explicitly referred to the objectivity/impartiality of their fact-checking organization ($d = 0.29$, 95% CI [.13, .46], $p = .001$, $k = 9$) and those that did not ($d = 0.29$, 95% CI [.23, .36], $p = .001$, $k = 21$); ($Q(1) = 0.01$, $p = .99$). Likewise, effect sizes did not vary ($Q(1) = 0.97$, $p = .32$) by whether fact-checking messages included a logo ($d = 0.37$, 95% CI [.19, .54], $p = .001$, $k = 7$) or not ($d = 0.27$, 95% CI [.21, .34], $p = .001$, $k = 23$).

Publication Bias

A common method to detect a publication bias uses *Kendall's Tau* to estimate the relationship between the standardized effects and their variance (Begg & Mazumdar, 1994). According to this estimate, there is a significant relationship between standardized effects and their variance ($Tau = .30$, $p = .02$), indicating some support for a potential file drawer problem. Further, a funnel plot (plots Cohen's d effect sizes on the x-axis against a measure of study size on the y-axis; standard errors) is a useful approach to detect bias (Egger, Smith, Schneider, & Minder, 1997), with degree of funnel plot asymmetry indicating a possible publication bias. As Figure 4 illustrates, larger samples tend to produce

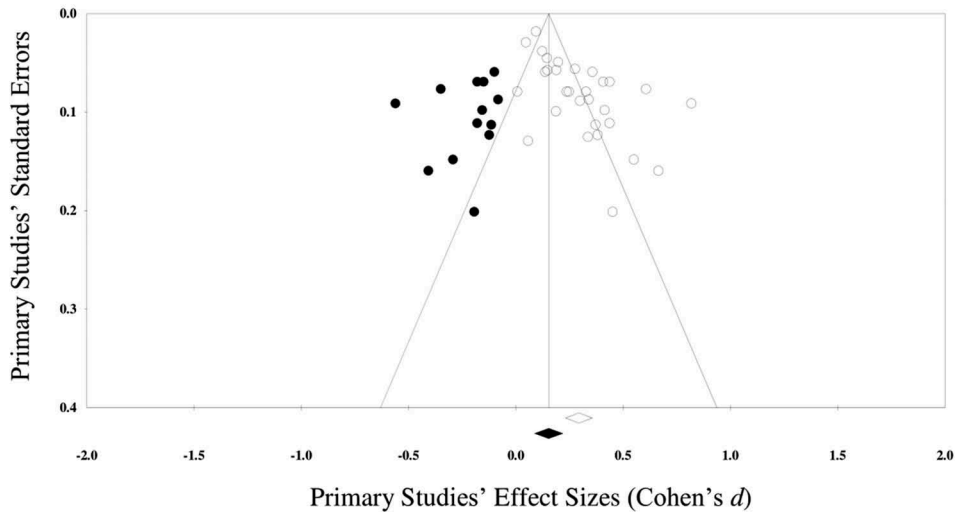


Figure 4. Funnel plot for the detection of a potential publication bias (open circles represent observed effect sizes and full circles represent imputed effect sizes).

somewhat weaker effects compared with studies that utilized smaller samples. Duval and Tweedie (2000) method of trim and fill was used to numerically assess this asymmetry. Following this approach, smaller studies causing plot asymmetry are removed and “true” averaged effect is estimated, while the center of the plot is filled with the omitted studies. Based on the trim and fill assessment in comprehensive meta-analysis, nine studies had to be trimmed, resulting in a slightly weaker average effect of fact-checking ($d = 0.21$, 95% CI [.14, .27]) than the one observed in the data. Likewise, when directly comparing published studies ($k = 16$) with unpublished studies ($k = 14$) within our sample, the analysis recorded a significant difference [$Q(1) = 11.03$, $p = .001$], indicating that unpublished studies tended to retrieve weaker effects ($d = 0.18$, 95% CI [.12, .24]), compared to published studies ($d = 0.39$, 95% CI [.28, .49]). In total, this provides considerable evidence for a publication bias in the fact-checking literature.

Discussion

In its brief history, the fact-checking enterprise has drawn praise and criticism due to its role as an unbiased referee in a highly polarized political environment (Marietta et al., 2015). When disputed realities are involved, any effort to correct misinformation and inform the public by utilizing facts can be expected to both have effects and to encounter resistance. Though fact-checking organizations have successfully become a standard part of political reporting (Graves et al., 2016), their growing prominence raises important questions regarding effectiveness. Hence, the question at the center of the current study is the following: *What is the ability of fact-checking messages to affect people's beliefs?*

There are two potential answers to this question offered in this meta-analysis. Striking a more optimistic tone, fact-checking messages positively affect beliefs, irrespective of political ideology, preexisting positions, context (campaign vs. routine), and whether it refutes the entire false statement or just parts of a statement. In fact, as the forest plot clearly demonstrates, though not all fact-checking attempts are equally

effective, there is no evidence of factual backfire (Nyhan & Reifler, 2010). As argued by Wood and Porter (2019), “citizens heed factual information, even when such information challenges their ideological commitments” (p. 1). Simply put, the beliefs of the average individual become more accurate and factually consistent, even after a single exposure to a fact-checking message. To be sure, effects are heterogeneous and various contingencies can be applied, but when compared to equivalent control conditions, exposure to fact-checking carries positive influence.

However, the results also raise substantial concerns. In line with the motivated reasoning literature (Kunda, 1990; Nir, 2011), the effects of fact-checking on beliefs are quite weak and gradually become negligible the more the study design resembles a real-world scenario of exposure to fact-checking. For instance, though fact-checking can be used to strengthen preexisting convictions, its credentials as a method to correct misinformation (i.e., counter-attitudinal fact-checking) are significantly limited. Likewise, even if journalists and elites often hail the independent and objective nature of fact-checking organizations (Fridkin et al., 2015; Nyhan & Reifler, 2015), the results point to visible differences between how Democrats/liberals and Republicans/conservatives interpret fact-checking information. Whereas Democrats/liberals are equally receptive to information that supports or contradicts their ideology, Republicans/conservatives are more eager to accept pro-attitudinal corrections and less likely to adopt ideologically inconsistent information. In agreement with previous research (Shin & Thorson, 2017), it seems that the quality of objectivity often ascribed to fact-checking organizations does not transcend political and ideological divides. Ostensibly, this finding may speak to inherent differences between Republicans/conservatives and Democrats/liberals, with the former being more likely to engage in biased processing. From a strategic communication point of view, however, these findings are perhaps indicative of an ongoing attempt to undermine fact-checking efforts that contradict party lines (Stencel, 2015). For example, President Trump – with an astonishingly poor record among professional fact-checkers (Graves, 2017) – has repeatedly dismissed discernible reality by labeling legacy media and the fact-checking of his statements as “fake news” (Jamieson & Taussig, 2017). Arguably, this aggressive approach to elite journalism and a growing disdain for fact-checking helps thwart criticism, making partisan bias among Republicans considerably worse.

In addition, fact-checking messages that included graphical elements tended to be less effective than fact-checking messages that did not integrate visuals. We initially expected graphical elements to facilitate corrections because they provide simple and straightforward summary of the fact-checking message, but the data did not support this line of reasoning. Rather, graphical elements appeared to interfere with the correction. It is difficult to speculate why the inclusion of visual elements tended to backfire, one potential cause has to do with motivated reasoning. As argued by Amazeen et al. (2018), truth scales can serve as a strong partisan cue, confirming a person’s pre-existing beliefs (either favorable or unfavorable) thus eliminating the motivation to centrally process the rest of the message.

Further, though misinformation is a constant problem, individuals are more likely to be exposed to misinformation or disinformation during election campaigns (Coddington et al., 2014). But it is during these times, when fact-checking is needed most, that motivated reasoning is at its peak (Miller & Conover, 2015), and so fact-checking is less effective. The results also question the ability of fact-checking organizations to increase public knowledge. In fact, what sets fact-checking messages apart from most other approaches to correction of misinformation is their emphasis on providing more context that can help individuals accurately evaluate political entities. The focus on the political process and public policies often necessitates the use of nuanced and complex language, which can have both favorable and unfavorable

consequences. As a whole, lexical complexity appears to detract from fact-checking efforts, whereas simpler and less sophisticated messages prove to be more effective.

Finally, given the inherent complexities of our political reality, the ratings of statements made by political entities are more likely to be in a shade of gray rather than completely true or false (PunditFact, 2018). For example, an hour-long State of the Union address can strike a variety of false notes along with some exaggerations, as well as generate completely accurate statements. Keeping in mind, however, that fact-checkers focus solely on factual statements, even a ruling of “mostly false” can increase the credibility of various ungrounded statements. After all, if a long speech was thoroughly fact-checked and opinion statements were not directly refuted, perhaps, it means that they are accurate (Pennycook & Rand, 2017). Interestingly, there is some evidence to suggest that even different fact-checking organizations find it difficult to agree on an ambiguous scoring range (e.g., “mostly false”) (Lim, 2018); thus, the limited efficacy of partial corrections can be the result of political discourse that is designed to be ambiguous rather than audience’s cognitive biases (e.g., implied truth effect). Practically speaking, this may suggest that fact-checking organizations need to carefully consider their units of analysis. For example, instead of focusing on entire speeches that are likely to result in equivocal verdicts (e.g., mostly true/mostly false), an alternative approach can focus on specific statements leading to more conclusive judgment (e.g., true/false).

The limited utility of fact-checking is also evident when directly contrasting the findings with results from recent meta-analyses that addressed the psychological efficacy of messages countering misinformation (Chan et al., 2017; Walter & Murphy, 2018) and ways to protect eyewitness memory against the misinformation effect (Blank & Launay, 2014). In comparison to the weak effects recorded in the current study, other meta-analyses observed strong (Blank & Launay, 2014; Chan et al., 2017) and moderate effects (Walter & Murphy, 2018). Presumably, these discrepancies can be attributed to the challenging context in which fact-checking organizations operate. In particular, while previous analyses have aggregated the effects of political misinformation together with topics that are less likely to induce resistance and partisanship (e.g., health, marketing), the weaker effects obtained in the current study can be potentially attributed to the value-laden contexts that studies of fact-checking typically address. This explanation is supported by the fact that when focusing exclusively on political topics, there is virtually no difference between the main effects recorded in Walter and Murphy ($r = .15$, 2018) and the current meta-analysis ($r = .15^{10}$).

At this point, it is important to situate the current results in the broader context of fact-checking as a journalistic practice. Though the current study focuses exclusively on the direct effects of political messages on citizens’ beliefs, fact-checking is not only about changing what citizens think – it is also, to a large extent, about holding political entities accountable by keeping false statements out of public discourse (Graves, 2016). According to this view, if political figures think that their statements will be subjected to robust scrutiny, they will be less likely to make, or repeat, false claims. Additionally, in line with the third-person effect, if politicians feel that their voters are likely to be affected by exposure to fact-checking messages, they, in turn, would attempt to change their rhetoric and actions (Cohen, Tsfat, & Sheaffer, 2008). As these scenarios indicate, fact-checking can be a powerful tool of social influence even if it has only limited efficacy when it comes to changing public opinion.

Limitations and Future Directions

The most obvious limitation associated with meta-analysis is its dependence on primary studies, which is exacerbated when the analyses are based on a relatively small number of

studies. In such cases, not all hypotheses can be adequately tested and the power to detect significant moderators is somewhat compromised. Likewise, the relatively small sample of studies has limited our ability to provide a more nuanced understanding of fact-checking. For instance, distinct constructs such as ideology (conservative vs. liberal) and partisanship (Republican vs. Democrat) were combined rather than analyzed separately. Further, the decision to focus on highly controlled experimental designs assisted in identifying the causal link between fact-checking and beliefs, but it also raises the question whether the current results can be generalized to more naturalistic settings. For example, does exposure to fact-checking messages produce the same effects when people are instructed to carefully read a message in a lab setting, as opposed to co-viewing a political debate in a living room with family and friends? What is the role of forced exposure to messages versus having the real-world choice of whether to consume a fact-check message (see Garrett, 2013)? Keeping in mind that extant literature lacks tests of such comparisons, these questions remain unanswered.

Further, the current analysis did not include empirical reports that tested the role played by fact-checking using correlational methods (e.g., Karlsson, Clerwall, & Nord, 2017). While including correlational studies, and treating methodology as a moderator, may increase the sample size, it also introduces bias into our results. Namely, experimental designs, where the effect of fact-checking is isolated, permit conclusions about whether the causal effect of fact-checking on beliefs. Correlational studies, however, permit conclusions about whether fact-checking is associated with beliefs, but cannot possibly speak to the effects of fact-checking. Considering the fact that correlational designs attempt to answer distinct questions, even if their inclusion is desirable in terms of comprehensiveness, conceptually and methodologically such syntheses are difficult to justify. Relatedly, the fact that only 27.6% of the studies in the sample employed authentic fact-checkers (the remaining studies used fictitious fact-checkers) hinders the external validity of the findings. Likewise, additional moderators such as issue-involvement and political efficacy are expected to provide a more fine-grained understanding of fact-checking; yet, such variables were largely ignored by primary studies.

As to be expected with a relatively young research program, there are numerous gaps to be filled in fact-checking research. First, future research should expand the palate of potential outcomes beyond beliefs and attitudes, to include effects on political knowledge, cynicism, efficacy, and information sharing. After all, if the primary goal of fact-checking organizations is to infuse public discourse with factual data, political knowledge or information seeking and sharing are perhaps better indicators of successful fact-checking. Second, while the number of fact-checkers around the world has more than tripled in the past few years (Stencel & Griffin, 2018) and currently fact-checking organizations can be found in 53 countries around the globe, the empirical literature focuses disproportionately on the United States. Hence, one promising stream for future research could advance a cross-cultural approach to fact-checking, contrasting well-established democracies with other types of political systems (see Van Aelst et al., 2017). Finally, given the struggle of social media and technology companies to keep up with the sophistication of viral hoaxes and disinformation campaigns, more attention should be given to understand the most effective ways in which fact-checking practices can be integrated into online platforms.

Conclusion

Only recently has the empirical study of political misinformation and fact-checking correction produced a critical mass of work that is worthy of meta-analytical review. This area of

study is at a critical juncture in its maturation process while continuing to increase in social significance as fake news generation and dissemination become more pervasive in our digital media environments. Although much remains to be uncovered about the influence of fact-checking, we hope that the current study can serve as a roadmap to continue exploring the challenging terrain of political misinformation and its correction.

Notes

1. The typical procedure in the control condition involved exposure to a message that included misinformation and then exposure to a measurement of message-relevant beliefs. Importantly, message design did not account for significant differences in effect sizes [$Q(1) = 0.24$, $p = .63$]; between-subjects $d = 0.28$, 95% CI [.21, .36], $k = 23$ vs. within-subjects $d = 0.32$, 95% CI [.19, .46], $k = 7$.
2. Though the original plan was to include both favorable and unfavorable fact-checking, the paucity of favorable fact-checking research would have prevented us from analyzing the data in a statistically meaningful way.
3. Issue favorability $d = 0.36$, 95% CI [.23, .49] vs. issue accuracy $d = 0.26$, 95% CI [.19, .33].
4. Economy $d = 0.14$, 95% CI [.08, .19] vs. education $d = 0.15$, 95% CI [-.13, .44] vs. healthcare $d = 0.28$, 95% CI [.15, .40] vs. national security $d = 0.26$, 95% CI [.06, .47].
5. Real-world issue $d = 0.30$, 95% CI [.23, .37] vs. fictional issue $d = 0.26$, 95% CI [.10, .42].
6. Interest group $d = 0.27$, 95% CI [.15, .38] vs. politician $d = 0.33$, 95% CI [.01, .43] vs. the press $d = 0.22$, 95% CI [.16, .28].
7. Due to the fact that some variables were coded at the subsample level, several moderators add up to more than 30 cases (see Table 1).
8. Due to the fact that lexical complexity cannot be meaningfully assessed for texts with less than 50 words, one study (i.e., Agadjanian et al., 2019) was not coded for this moderator.
9. In terms of research design characteristics, there were no significant differences between real-world fact-checkers and fictitious fact-checkers ($Q(1) = 2.41$, $p = .12$); fictional misinformation and real-world misinformation ($Q(1) = 0.29$, $p = .87$).
10. Transformed from Cohen's $d = 0.29$.

Disclosure statement

No potential conflict of interest was reported by the authors.

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An asterisk precedes references included in the meta-analysis.

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